2.7 **[2,0점]** ,

.

(4)

$$dp_{vap} = -\rho_{vap} g dh, \tag{26}$$

$$\rho_{vap} = \frac{\mu_{H_2O} p_{vap}}{RT}, \tag{27}$$

(15) .

$$p_{vap}(0) = \varphi p_{vap0}, \tag{28}$$

H'

$$p_{vap}(h) = p_{vap}. \tag{29}$$

(25) (26) (15) (28) (29)

$$\ln \frac{p_{vap}}{p_{vap0}} = \ln \varphi + \frac{\mu_{H_2O} g}{\beta R} \ln \frac{T}{T_0}. \tag{30}$$

$$\ln \frac{P_{vap}}{P_{vap0}} = a + b \ln \frac{T}{T_0}, \tag{31}$$

(30) H'

$$T(H') = T_0 \exp\left(\frac{a - \ln \varphi}{\frac{\mu_{H_2O} g}{\beta R} - b}\right). \tag{32}$$

(15)

$$H' = \frac{T_0 - T(H')}{\beta} = \frac{T_0}{\beta}\left(1 - \exp\left(\frac{a - \ln \varphi}{\frac{\mu_{H_2O} g}{\beta R} - b}\right)\right) = 2.55 \cdot 10^3 \text{ m}. \tag{33}$$

2.8 **[0,6점]** (33)

$$H' = H_0, \tag{34}$$

$$\varphi_{min} = \left(1 - \frac{\beta H_0}{T_0}\right)^{b - \frac{\mu_{H_2O} g}{\beta R}} \exp a = 0.119. \tag{35}$$

	평가 항목	배점	
1.1	(1) $p_0 S = Mg$	0,4	**1,0**
	(2) $S = 4\pi R_E^2$	0,2	
	(3) $M = \frac{4\pi p_0 R_E^2}{g}$	0,2	
	$M = 5.32 \cdot 10^{18}$ kg	0,2	
1.2	(4) $dp = -\rho g dh$	0,2	**1,0**
	(5) $\rho = \frac{\mu_{air} p}{RT_0}$	0,2	
	(6) $p(h) = p_0 \exp\left(-\frac{\mu_{air} g}{RT_0} h\right)$	0,4	
	$p(H) = 85.0 \cdot 10^3$ Pa	0,2	
1.3	(8) $\delta Q = \frac{M}{\mu_{air}} \frac{\gamma R}{\gamma - 1} \Delta T$	0,2	**0,6**
	$\delta Q = 5.33 \cdot 10^{21}$ J	0,2	
	(9) $\gamma = 7/5$ $C_P = 7/2R$	0,2	
1.4	(10) $\delta Q = \alpha \pi R_E^2 \tau$	0,2	**0,6**
	(11) $\tau = \frac{M}{\alpha \pi R_E^2 \mu_{air}} \frac{\gamma R \Delta T}{\gamma - 1}$	0,2	
	$\tau = 30.3 \cdot 10^3$ s	0,2	